

There are multiple advantages to this approach:

- The class hides all the details of pygame's rendering of text, so the user of this class never needs to know what pygame-specific calls are needed to show text.
- Each `SimpleText` object remembers the window that it draws into, the location where the text should be placed, and the text color. Therefore, you only need to specify these values once, when you instantiate a `SimpleText` object, typically before the main loop starts.
- Each `SimpleText` object is also optimized to remember both the text that it was last told to draw and the image (`self.textSurface`) that it made from the current text. It only needs to render a new surface when the text changes.
- To show multiple pieces of text in a window, you only need to instantiate multiple `SimpleText` objects. This is a key concept of object-oriented programming.

Demo Ball with SimpleText and SimpleButton

To cap this off, we'll modify [Listing 6-2](#) to use the `SimpleText` and `SimpleButton` classes. The updated program in [Listing 6-7](#) keeps track of the number of times it goes through the main loop and reports that information at the top of the window. Clicking the Restart button resets the counter.

File:

PygameDemo8_SimpleTextDisplay/Main_BallTextAndButton.py

```
# pygame demo 8 - SimpleText, SimpleButton, and Ball

# 1 - Import packages
import pygame
from pygame.locals import *
import sys
import random

❶ from Ball import * # bring in the Ball class code
from SimpleText import *
from SimpleButton import *
```